

S1

Baseline

All factors neutral;
control group

S2

High Low-Level Demand

Tests vehicle
upgrade logic

S3

Geographic Imbalance

Tests fleet
relocation needs

S4

High Order Load

Tests order
rejection logic

S5

Tight Relo Budget

Tests resource
constraints

GENERAL PARAMETERS

Parameter	S1	S2	S3	S4	S5
n_S — Stations	5	5	5	5	5
n_C — Cars	10	10	10	10	10
n_L — Car Levels	3	3	3	3	3
n_K — Orders	20	20	20	40	20
n_D — Planning Days	5	5	5	5	5
B — Relocation Budget (min)	600	600	600	600	150

FLEET COMPOSITION & PRICING

Parameter	S1	S2	S3	S4	S5
Level-1 Cars	4	4	4	4	4
Level-2 Cars	4	4	4	4	4
Level-3 Cars	2	2	2	2	2
Hourly Rate — L1	100	100	100	100	100
Hourly Rate — L2	200	200	200	200	200
Hourly Rate — L3	500	500	500	500	500

ORDER DEMAND DISTRIBUTION

Parameter	S1	S2	S3	S4	S5
Order Level — L1	33%	70%	33%	33%	33%
Order Level — L2	33%	20%	33%	33%	33%
Order Level — L3	34%	10%	34%	34%	34%
Pick-up Station	Uniform	Uniform	Stn 1: 60%	Uniform	Uniform
Return Station	Uniform	Uniform	Stn 5: 60%	Uniform	Uniform

Parameter	S1	S2	S3	S4	S5
Rental Duration	1-48h	1-48h	1-48h	1-48h	1-48h

Blue values indicate parameters modified from the S1 Baseline to isolate specific operational challenges.